1. Problem I observed

During the Week 1 component scan, one thing became clear to me. My architecture already included a separate memory extraction and admission stage, but I was still not confident about how admission decisions should actually be made. I understood why every extracted piece of information should not become a long-term memory, but I did not have a concrete policy for deciding what should be stored and what should be discarded.

If admission decisions are too permissive, the memory store gradually fills with duplicate, temporary or low-value information. On the other hand, if the admission policy is too strict, useful memories may never be stored. Simply extracting memories from a conversation is therefore not enough; the system also needs a reliable way to decide whether each candidate memory deserves a place in long-term storage.

This uncertainty was also reflected in my D1 open questions, which is why I selected **Memory Extraction and Admission** as one of the three components for deeper validation.

2. Mechanism that stood out

The paper *Adaptive Memory Admission Control (A-MAC)* gave me a much clearer way to think about admission decisions.

Instead of treating admission as a simple filtering step, it frames it as a structured decision-making process. Rather than relying on a single rule, every candidate memory is evaluated using multiple signals before it is stored. These signals include the future usefulness of the memory, confidence in the extracted information, semantic novelty, temporal recency and the type of memory being stored.

What I liked most about this approach is that it balances lightweight rule-based checks with limited LLM reasoning instead of depending entirely on the LLM for every admission decision. That makes the approach much more practical for a production system where cost and latency also matter.

The paper also points out an important limitation of relying only on utility or novelty scores. A memory may appear useful, recent and unique while still being factually incorrect. That observation reminded me that admission policies should not depend on a single metric, but should combine multiple signals before making a storage decision.

3. Evidence from the paper vs. my interpretation

The paper provides experimental evidence showing that a structured admission policy produces a cleaner and more useful memory store than storing everything that is extracted from a conversation. It also demonstrates that combining inexpensive rule-based signals with selective LLM evaluation can reduce unnecessary memory creation while keeping computational cost manageable.

From those results, I am confident that introducing a better admission policy is a worthwhile improvement for my system.

However, one design choice is my own interpretation rather than something directly taken from the paper. I don’t want to copy the complete A-MAC framework into my baseline implementation. Instead, I plan to use the central idea behind it, a structured multi-signal admission policy and adapt it to the simpler architecture of my conversational memory system. This keeps the baseline implementation manageable while still addressing one of the biggest design gaps identified during Week 1.

4. Proposed design amendment

Based on this deep dive, I plan to keep the overall architecture identified during the Week 1 scan but improve the admission stage with a structured decision policy.

Instead of relying on a few simple checks, every candidate memory will be evaluated using multiple signals before it is stored. These signals will include confidence in the extracted information, semantic novelty, recency, memory type and an estimate of future usefulness. The final admission decision will be made by considering these signals together rather than depending on a single rule.

At the same time, I do not plan to implement the complete A-MAC framework in the baseline system. Some parts of the paper introduce additional complexity that is unnecessary for the first implementation. My goal is to adopt the core design idea while keeping the admission pipeline simple enough to build, test and improve in later iterations.

This design change directly addresses one of the biggest uncertainties identified during Week 1 and provides a much clearer strategy for deciding what should become long-term memory.

5. Risks and limitations

Although I decided to adopt this idea, there are still a few limitations that we should keep in mind.

The first limitation is cost. Estimating the future usefulness of every candidate memory may require an additional LLM call, which increases both latency and inference cost. The paper reduces this by combining lightweight rule-based checks with selective LLM reasoning, and I plan to follow a similar approach instead of relying entirely on the LLM.

Another limitation is that admission quality depends on the quality of the extracted candidate memories. If the extraction stage produces incorrect or incomplete memories, even a good admission policy may still make poor decisions. This means admission should be viewed as one part of the overall memory pipeline rather than a complete solution on its own.

Finally, the exact thresholds and weighting of different admission signals will require experimentation. The paper provides a useful direction, but those values should be validated within my own conversational memory system.

6. What would make me change this decision?

At the moment, I believe introducing a structured admission policy is the right decision for the baseline implementation.

However, I would reconsider this decision if experiments show that the additional admission step provides little improvement over a simpler rule-based approach while noticeably increasing cost or latency. I would also revisit the design if another admission strategy consistently produced better memory quality with less implementation complexity.

Conclusion

This deep dive helped answer one of the biggest open questions that remained after the Week 1 component scan. I already understood why memory extraction should be separated from memory storage, but I was still unsure how admission decisions should actually be made. The A-MAC paper filled that gap by providing a practical and structured admission strategy that balances multiple signals instead of relying on a single rule.

Rather than adopting the complete framework, I plan to incorporate its central idea into the baseline conversational memory system while keeping the implementation simple enough for the first version. I think this approach gives a good balance between improving memory quality and keeping the architecture practical to implement.

Proposed memory admission pipeline for the baseline conversational memory system.